Benson (Zhen Cong) Chen

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AI Engineer specializing in low-level GPU optimization for machine learning system across mobile, desktop, and safety-critical environments. Llama.cpp contributor with cross-vendor GPU experience (Qualcomm, Nvidia, Apple) and a track record of measurable wins—1.5× Qwen3 inference on mobile GPUs, 60% runtime reduction on GEMM kernels, and BitNet ternary-quantization support across heterogeneous GPUs.

Skills & Competencies

- Programming languages: C, C++, Python, Java
- ML API: Vulkan, CUDA, Metal, PyTorch
- LLM: Llama.cpp, Safety Critical Environment, Extreme Quantization, GPU Kernel Optimization, AI Inference, Edge Deployment

Work Experience

AI RESEARCH ENGINEER (EDGE INFERENCE) TETHER.IO JAN 2026 – PRESENT

- Develop and optimize kernels, quantization workflows, and fine-tuning systems for LLMs/SLMs across mobile and desktop GPUs.
- Optimzed Qwen3 model on Andreno 840 GPU (Samsung S26 Ultra) to achieve 2x speed up compared to previous version.
- Added support for BitNet Heterogeneous GPU inference in llama.cpp by enabling ternary quantization in llama.cpp with GPU acceleration using Vulkan and Metal.
- Authored a research paper documenting the implementation of BitNet LoRA Fine-Tuning using Heterogeneous GPU.

ML SOFTWARE DEVELOPER (MODEL INFERENCE) COREAVI (LYNX) JAN 2024 – DEC 2025

- Performed benchmarking and optimization on GEMM operation for specific GPU and reduced application runtime by 80% for specific input size using optimization techniques such as memory tiling, register tiling, and thread coarsening.
- Built data processing pipeline for internal LLM tool.
- Enabled efficient linear quantization for matrix multiplication and improved computational performance.
- Automated software testing by leveraging Jenkins and Python script.
- Optimized GPU computation for safety-critical machine learning operations such as convolution.

ML SOFTWARE DEVELOPER (MODEL INFERENCE) COREAVI (LYNX) MAY 2023 – AUG 2023

- Developed machine learning engine in a safety-critical environment.
- Implemented testing library for ML inference engine to thoroughly test each ML operation for various inputs using PyTorch.
- Created debugging tools, improving developer productivity and troubleshooting accuracy.
- Built Resnet18, VGG16, VGG18, and other machine learning models using both PyTorch and native ML inference engine.
- Debugged large codebase, resolving issues and improving system reliability and performance.
- Refined NNEF compiler, ensuring accurate parsing and adherence to specifications.

COMPILER DEVELOPER HUAWEI COMPILER LAB SEP 2021 – DEC 2021

- Created an API for machine learning model to perform prediction for LLVM IR using SSH and network sockets.
- Generated and preprocessed training data for machine learning model.
- Added features on performance analysis infrastructure to improve ML model performance.
- Generated automated tests for compiler.

Education

THE UNIVERSITY OF TEXAS AT AUSTIN PRESENT

- Master of Computer Science, Candidate

UNIVERSITY OF WATERLOO 2023

- Bachelor of Computer Science (Co-operative Program), Dean's Honour